

Apply to be a Science Corps Fellow - Entry #3341

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Indian
When did you or do you expect to complete your PhD?
I expect to defend my thesis by April-May 2025
Please tell us briefly about your university education (institution, city, country, year)
I completed my undergraduate degree in Microbiology, Chemistry, and Zoology at St. Joseph's College, Bengaluru, India, in 2015. In that same year, I qualified and joined the integrated Master 's-PhD program at the National Centre for Biological Sciences, Bengaluru, India. Here, I pursued my master's and PhD degrees in the field of Evolutionary Biology in Krushnamegh Kunte's lab using butterflies as a model system. During my PhD, I studied the effects of natural and sexual selection on wing colour variation (an important visual signal) in butterflies. I have submitted my thesis and awaiting my defense.
Why would you like to be a Science Corps Fellow?
Science is omnipresent, and I believe anyone can understand and speak the language of science as it is universal and does not discriminate. However, inculcating a scientific temperament is often overlooked in school curricula. During my schooling, my teachers' primary motivation was to ensure an understanding of scientific concepts rather than the process of scientific discovery. Later, during my undergraduate education, laboratory practicals provided some hands-on experience, shifting my perspective from merely learning science to actively doing it. Then my PhD showed me that answering one question often leads to many more, reinforcing that science is a continuous and evolving process. It was during my PhD that I realized science is rarely a straightforward path from problem to solution; rather, it is a journey filled with trial, error, and refinement. I believe these aspects of science should be highlighted throughout the learning process, not just at the stage of active research. During my PhD work on colour pattern evolution in butterflies, I encountered and overcame numerous challenges, from troubleshooting why caterpillars were dying on their host plants to debugging unexpected issues in my statistical code. I gained valuable experience in failing multiple times before finally arriving at an answer. This

taught me the value of persistence—a skill that should be sown early during education. However, I was truly exposed to this only in my PhD when I had my own projects and my own questions to ask. This made me realise that fostering an inquisitive mindset needs to begin early on during the educational life cycle of a child. As a Science Corps Fellow, I hope to implement a hands-on, inquiry-driven approach to biology, where students can identify simple, yet meaningful questions based on their lessons, design experiments or analyse information, and critically evaluate their findings. Simultaneously, I would like to stress on the benefits of adopting a scientific temperament in all aspects of life. Additionally, I am passionate about making science accessible, and my experience in insect ecology and evolutionary biology offers a unique opportunity to enable students to engage with the community through citizen-science activities such as nature walks and biodiversity monitoring. I would also be excited to facilitate scientific discussions and encourage independent thinking through activities such as dissecting a research paper. This fellowship would provide valuable experience in science communication and teaching, a crucial aspect that can supplement my academic interests. It would offer an opportunity to integrate research and education, extending my scientific reach.

I also hope to gain exposure to a rich community in the host institute and to be able to share and exchange ideas to grow as a scientist and educator. This experience would enhance my communication skills—both scientific and otherwise—and ability to engage with students.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

I have always been passionate about science outreach and education. During my PhD, I participated in campus-wide Open Days where we set up exhibits pertaining to our work or broader concepts and engaged with school children of different ages and socio-economic backgrounds. For instance, my labmates and I designed an interactive game to highlight evolutionary concepts such as aposematism, mimicry, and fitness. In this game, students had to imagine they were hungry, inexperienced birds looking for their next meal. The game consisted of identifying toxic from non-toxic butterflies through two phases, a learning phase and a test phase. Their choices ultimately determined whether they survived and made it to the end, i.e., did they have greater fitness compared to their fellow birds. Designing simple games like this sparked my interest in finding alternate ways to showcase scientific ideas. Additionally, in my efforts to make science accessible, I also delivered a public talk during a Butterfly Awareness Month on colour diversity and evolution in butterflies. I have also participated in and co-organised two sessions of a scientific discussion series called Sympotein. In this series, we would deep dive into a topic outside of our area of expertise and share it with a broad yet scientific audience in a “popular science” format. This involved thorough researching, preparing, and presenting talks on topics unrelated to our thesis work. These experiences were key in sparking my interest in science education and outreach.

In a more formal setting, I also curated and delivered a lecture for undergraduate students as part of a larger lecture series on feedback in biology, where I spoke about eco-evolutionary feedback mechanisms. This gave me the experience to design a short two-hour module from scratch, where I had the creative and academic freedom to explore and improve my method of teaching. I have also delivered a research talk on my work to master's students at my alma mater. Apart from speaking to larger groups, I also have mentored students 1-on-1 within the lab. I mentored master's thesis students and summer interns for their projects, which included demonstrating skills such as insect handling and rearing, basic R programming, and pertinent lab techniques. Through this experience, I helped students with specific problems pertaining to their research work and worked together with them to troubleshoot their problems.

Each of these experiences has helped me grow not just as a scientist but as an educator. They have equipped me

with some of the skills required to be an effective teacher, either to a broad or specialized audience. However, most of these talks and outreach activities were isolated events, scattered through my PhD journey without a formal structure and everyday learning objectives and assessments. This fellowship would be key in providing this formal avenue to refine my mentorship skills and further develop in an advisory role.

Where do you imagine yourself professionally in the future? Check all that apply.

Scientist in academia

In science education

In science outreach

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

At this stage in my career, I am inspired to explore and carry out research further. Therefore, my short-term goal is to continue my research trajectory as a postdoctoral fellow in evolutionary biology. During this phase, I would like to explore novel areas and develop new technical skills. I also hope to improve my teaching and mentoring skills. The Science Corps Fellowship would be invaluable in honing my science communication and education skills, which I can apply during my postdoctoral phase.

I believe that a mentor plays a crucial role in providing the right tools and guidance to students. Thus, in the long term, I envision being a mentor either as an educator or facilitator of science. This could be at the advanced research level (by mentoring my own lab members) or at the foundational school and college levels. I hope to equip students with the skills and knowledge necessary for academia or other career paths by sampling from the experiences gained during my PhD and future postdoc. While I have an open mind about the exact job I will be doing in 10 years' time, my vision is to integrate research and teaching. I want to inculcate a scientific temperament in students such that they solve problems using the principles of the scientific method even beyond academic settings. The needs of students and society rapidly evolve, and I think scientists and educators must adapt accordingly. While I have some formal research experience, this fellowship would be an essential first step for me in a formal educational environment. I believe research scientists gain a lot from making their science more accessible, and one needs to be able to communicate effectively and engagingly to inspire future generations of scientists to come.

Please upload your CV (.pdf)

 [BhavyaDharmaraaj_CV_2025.pdf](#)

How did you find out about Science Corps?

An email list in your institution

Referred by a previous fellow

[Science Corps](#)